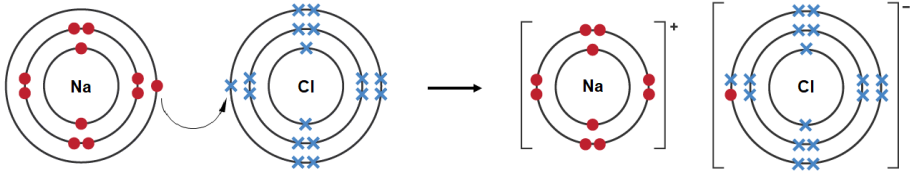


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)	I	 electron transfer and correct electronic structure of ions (1) charge on both ions (1)		2		2		
			II	Na ⁺ ion smaller than Cs ⁺ ion (so fewer Cl ⁻ ions can pack around it)	1			1		
	(b)	(i)		yellow (1)	1					1
				$\text{Ag}^+(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{AgI}(\text{s})$ (1)		1		2		
				ignore state symbols						
		(ii)		iodine (causes the brown colouration) (1)	1					1
				(bromine is an) oxidising agent (1)		1		2		
	(c)			add any suitable sulfate solution e.g. sodium sulfate (1)			1			
				no change with Mg ²⁺ and white precipitate with Ba ²⁺ (1)	1			2		
				or						
				add sodium hydroxide solution (1)						
				white precipitate with Mg ²⁺ and no change with Ba ²⁺ (1)						
				Question 9 total	10	4	1	15	0	4